Escape speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain escape in an ideal gravitational-energy model.

SUCCESS CRITERIA - I can...

1. State the zero-at-infinity energy convention.
2. Distinguish escape speed from circular-orbit speed.
3. Name neglected effects in the ideal calculation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “escape speed” and “orbit” to explain your model or result. Sentence starters are optional.

escape speed: minimum ideal speed to leave without further thrust

orbit: a path around another body under gravity

radius: distance from the center to the edge of a circle or sphere

01 Bring back an earlier idea

Total kinetic energy changes from 15 J to 12 J in an isolated collision. Is it elastic? Where could the difference go?

Word help: kinetic energy = energy associated with motion

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $v_{escape} = \sqrt{(2\ GM/r)}$; $v_{circular} = \sqrt{(GM/r)}$. Ideal spherical body; ignore atmosphere/rotation. Minimum escape reaches infinity at zero speed with no further thrust.

Worked reasoning: For a toy planet with $GM=50\ m^3/s^2$ and radius 4 m, escape speed is $\sqrt{(2\ GM/R)}=5$ m/s. At that limiting speed the ideal final speed at infinity is zero.

Your turn: Draw initial kinetic and gravitational-potential-energy bars whose total is zero.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Look and explain

Does escaping mean that gravity turns off at the launch point?
Explain what the speed allows in an energy model.

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Marzano check - circle: ① ② ③

04 Build the solution

A model needs 50 J to reach infinity. A projectile begins with 60 J kinetic energy. What energy is left at infinity, neglecting losses?

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Escape speed

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

Why is an ideal escape-speed calculation not a complete rocket-launch plan?

Word help: ideal = using stated simplifying conditions

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3

06 Choose a method

Lift a 6 kg object by 8 m.
Use g=10 N/kg.
Find the change in gravitational potential energy.

Word help: potential energy = energy associated with the configuration of interacting objects

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

An object has less energy than required to escape and receives no further thrust. Can this ideal model expect it to escape?

Word help: ideal = using stated simplifying conditions

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

The same toy planet's circular orbital speed at its surface is supplied as 3.54 m/s.
Compare it with escape speed 5 m/s and explain their different purposes.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3